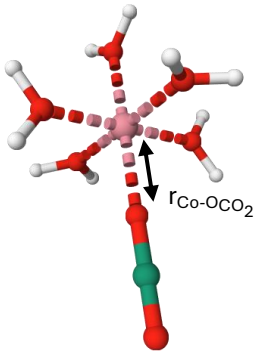
# Open-shell SQD for transition-metal chemistry

*Sample-based quantum diagonalization on quantum hardware captures charge-transfer physics and solvent effects*

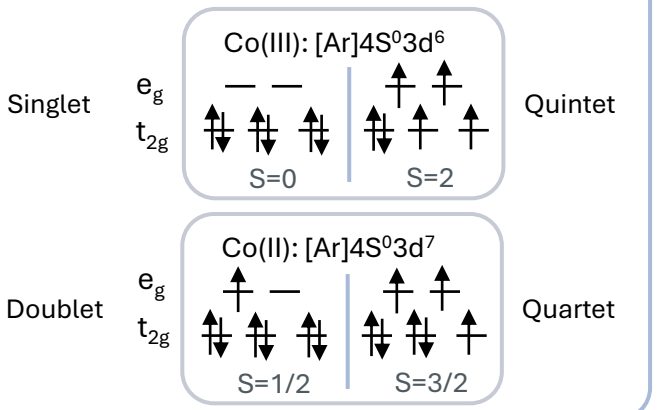
## TARGET SYSTEM

$[Co(H_2O)_5CO_2]^{2+/3+}$

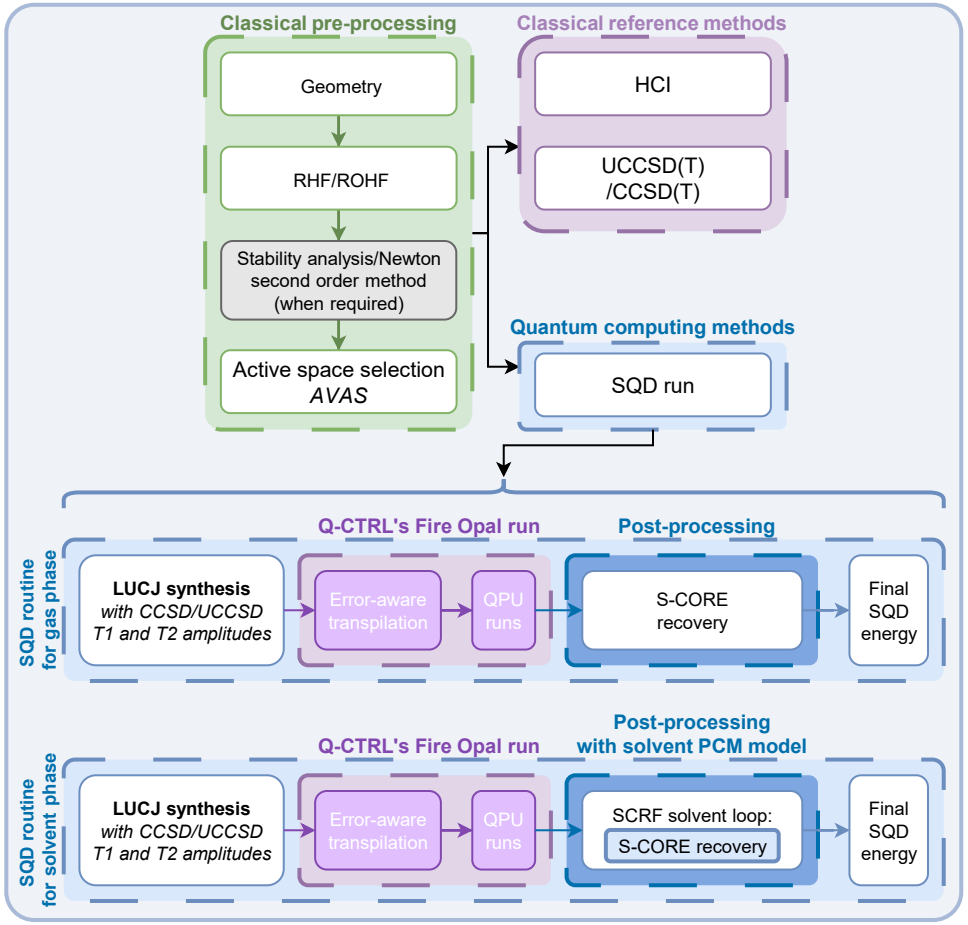



C H O Co

**Two oxidation states • Four spin multiplicities**
**Octahedral coordination**




## OVERALL WORKFLOW




## KEY RESULTS

### Benchmarking accuracy

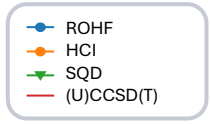
Open-shell SQD reproduces HCI benchmark energies for transition-metal system in gas and solvent phases. Mean error ± standard deviation (mHa):

| Charge 3+ | (22e, 17o) | (24e, 19o) | (28e, 23o) | (30e, 25o)(*) |
|---|---|---|---|---|
| Gas phase | 0.81±0.40 | 1.18±0.75 | 2.52±0.51 | 2.59±2.03 |
| Solvent phase | 0.54±0.31 | 0.55±0.41 | 0.98±0.61 | 5.31±1.77 |

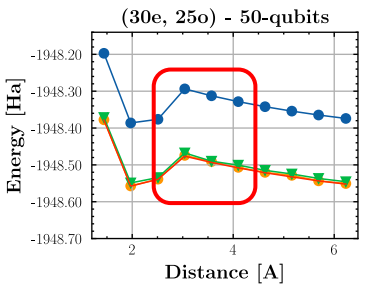
(*) Larger deviations in energy observed due to the limited post-processing time. More info in SI 3.

### Charge transfer physics

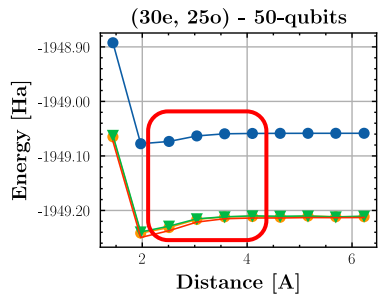
**Co(III) quintet** in gas phase shows an internal charge-transfer; Implicit solvent stabilizes the dissociation of neutral $CO_2$.




Gas




Solvent






**Open-shell transition-metal complex**

$[Co(H_2O)_5CO_2]^{2+/3+}$

**Open-shell SQD on quantum hardware**

Noise-robust sampling + classical post-processing

**Classical methods benchmark & charge-transfer check-up**

ROHF/UCCSD(T)/HCI runs
Population analysis

**IEF-PCM solvent**

Implicit solvent model favours dissociation of neutral $CO_2$ vs. $CO_2^+$ in gas phase